

Identifiability-Aware Calibration of the Stanford RRAM Model: Out-of-Sample Generalization and Deposition-Resolved Parameter Extraction for Sputter-Deposited TaO_x Devices

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Abstract—Compact-model calibration for resistive random-access memory (RRAM) is usually judged by the overlay between simulated and measured I - V curves, but a good overlay establishes neither that the fitted parameters generalize beyond the single curve used for fitting nor that they are uniquely determined. We present a reproducible calibration flow for the Stanford-PKU RRAM model that combines conduction-regime classification, regime-conditioned priors, mechanism-aware loss weighting, HSPICE-in-the-loop Bayesian optimization, and—as its primary deliverables—an *out-of-sample generalization test* and a *cycle-bootstrap identifiability audit*. Applied uniformly to twelve sputter-deposited TaO_x conditions spanning 20–35% oxygen and 75–250 W power, the flow reproduces the measured DC butterfly curves with mean log-current RMSE of 0.530 (SET) and 0.761 (RESET) decades. Crucially, when the model fitted to a single representative (medoid) cycle is scored against *every other* measured cycle of the same device, the held-out median RMSE (0.542 SET, 0.800 RESET decades) is statistically indistinguishable from the in-sample value: the calibration is a property of the device, not an artifact of the chosen target. Across 200 cluster-bootstrap resamples per condition the median parameter coefficient of variation is 0.448; only the branch voltage scales are repeatable (CV $\approx$ 0.28), whereas series resistance and velocity prefactors are unstable (CV $>$ 1). Consequently, only the repeatable, identifiable combinations carry deposition information: the branch voltage scales vary systematically across the design of experiments, while series resistance—although locally Fisher-identifiable—spreads over two orders of magnitude and cannot support a response surface. Finally, the post-RESET high-resistance state is Schottky-dominant in 92.5–100% of cycles for all twelve conditions, with physically admissible dynamic permittivities (4.7–32), a mechanism outside the Stanford bulk current law that bounds achievable fidelity. Generalization, identifiability, and mechanism validity should therefore be reported together as the deliverable of RRAM compact-model calibration.

Index Terms—RRAM, TaO_x, compact model, Stanford model, identifiability, Bayesian optimization, out-of-sample validation, conduction mechanisms, variability.

I. INTRODUCTION

RESISTIVE random-access memory (RRAM) based on metal-oxide filamentary switching is a leading candidate for embedded nonvolatile memory and in-memory computing [1]–[3]. Circuit- and architecture-level exploration of RRAM systems relies on compact models, of which the Stanford-PKU

model [4], [5] is among the most widely used: it captures bipolar filamentary switching through a single gap state variable, a sinh-type tunneling current, and a temperature-accelerated gap growth/rupture law. Alternative physics-based and behavioral compact models exist [6]–[8], but all share the same practical difficulty addressed here.

A practical difficulty hides inside every reported “calibrated” parameter set. The Stanford model exposes more than twenty parameters, while a DC butterfly sweep constrains a handful of effective degrees of freedom: two resistance branches, two switching thresholds, and the curvature of each branch. Such over-parameterized models are *sloppy* in the sense of Transtrum, Gutenkunst, and co-workers [9], [10]: the data constrain a few stiff parameter combinations and leave the remaining directions to the optimizer’s discretion. Two laboratories fitting the same measurement can therefore publish different “extracted” activation energies or gap geometries, each with an excellent overlay plot. Without an identifiability analysis [11], [12], the reader cannot tell which numbers are measurements and which are artifacts of the prior or the search path.

This difficulty has two distinct consequences that the literature usually conflates. The first is *generalization*: does a parameter set fitted to one representative curve also describe the other measured cycles of the same device, or has the optimizer merely memorized one trace? The second is *identifiability*: even granting a good and generalizing fit, which individual parameters are pinned down by the data and which float? We treat both as first-class, quantitatively reported deliverables of compact-model extraction rather than afterthoughts.

We present an automated, physics-guided pipeline that fits the Stanford model to measured TaO_x DC switching data and accompanies every fitted vector with (i) an out-of-sample test that scores the single-cycle calibration against all remaining measured cycles, and (ii) cycle-level cluster-bootstrap confidence intervals, with a Fisher/Cramér–Rao report retained as a *local corroborating* diagnostic. The pipeline is applied uniformly to twelve TaO_x deposition conditions spanning a face-centered factorial design in oxygen flow and sputter power, so the same evidential standard applies to every extracted parameter set and so the identifiable parameter combinations can be tracked across the design of experiments (DOE).

The specific contributions are:

- 1) *Physics as a front-end diagnostic*. Every resistance-

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state segment of the representative switching cycle is classified into ohmic, Poole–Frenkel (PF), Schottky, or Fowler–Nordheim (FN) conduction using BIC model selection plus a dynamic-permittivity admissibility test. The regime map conditions the prior windows and loss weights and documents where the Stanford current law is structurally unable to follow the data.

- 2) *Out-of-sample generalization as a deliverable.* Because the full measured-cycle ensemble is retained, the model fitted to a single medoid cycle is scored against every other measured cycle of the same device. We show the held-out RMSE matches the in-sample value to within 0.04 decades on average—direct evidence that the calibration generalizes rather than overfitting the chosen target.
- 3) *Bootstrap-led operational identifiability.* Cycle-level cluster-bootstrap intervals are the primary identifiability evidence; Fisher analysis is reported as a local cross-check. This separates coordinates that are both locally constrained *and* repeatable from those that merely look locally constrained.
- 4) *Deposition-resolved extraction across the DOE.* Tracking the extracted parameters across twelve conditions, we show that only the repeatable, identifiable combinations (branch voltage scales) carry deposition information, while a locally identifiable but bootstrap-unstable coordinate (series resistance) spreads over two decades and cannot support a response surface. The post-RESET HRS is additionally Schottky-limited—outside the Stanford bulk law—in nearly all cycles of all conditions, with physically admissible interface permittivities.

Section II describes the devices and measurement design; Section III summarizes the extraction pipeline; Section IV reports regimes, fits, generalization, identifiability, ablations, and deposition-resolved trends; Section V discusses implications; and Section VI concludes.

II. DEVICES AND MEASUREMENT DESIGN

The devices are sputter-deposited TaO_x cross-point RRAM cells (4 μm feature size) fabricated and characterized as part of an ongoing deposition-condition study [19], [20]; the asymmetric Ta-oxide bilayer switching stack follows the now-standard TaO_x device family [15], [21]. Twelve sample conditions (S1–S12) span a face-centered factorial design over oxygen percentage in the sputter ambient and RF sputter power, with four replicated center points (Table I). This design supports response-surface analysis of extracted parameters versus deposition conditions, which we exploit in Section IV-F.

Bipolar DC switching was measured with a Keysight B1500 semiconductor parameter analyzer; raw data are CSV exports containing one or more SET, RESET, or mixed sweep blocks per file. The fitting current compliance is $I_{cc} = 5 \times 10^{-4}$ A; because the Stanford Verilog-A model contains no instrument compliance circuit, simulated currents are clamped to this magnitude after each transient. An automated quality screen (minimum point count, voltage span, peak-current window, dynamic range, and SET+RESET completeness) rejects shorted,

TABLE I
DEPOSITION DESIGN OF EXPERIMENTS. CENTER POINT (27.5%, 162.5 W) IS REPLICATED FOUR TIMES (S9–S12).

Sample	O ₂ (%)	Power (W)
S1	20	75
S2	20	250
S3	35	75
S4	35	250
S5	20	162.5
S6	35	162.5
S7	27.5	75
S8	27.5	250
S9–S12	27.5	162.5

open, incomplete, or noise-floor sweeps before any fitting. For the representative S1 device, for example, all 40 measured cycles pass the screen with a mean quality score of 8.9/10.

III. EXTRACTION METHODOLOGY

The pipeline comprises eight stages; this section summarizes each stage and its design rationale. A complete stage-by-stage specification, including the prior table and all configured constants, is provided in the released implementation.

A. Representative Medoid Cycle

For each condition, the fitted device is selected by ranked criteria (number of good cycles, mean quality score, good-cycle fraction), which prevents the fitting target from being chosen by visual preference. Programming segments are isolated per cycle and branch by splitting the voltage trace at direction reversals. All cycles are interpolated in $\log_{10} |I|$ onto a common voltage grid, and the cycle that minimizes the median absolute deviation from the branch-wise median log-current,

$$s_{c,b} = \text{median}_v |\log_{10} |I_{c,b}(v)| - \text{median}_{c'} \log_{10} |I_{c',b}(v)||, \quad (1)$$

averaged over both branches, is retained as the *medoid* fitting target. Anchoring the point estimate to one measured sweep—rather than a synthetic pointwise median—preserves the measured hysteresis ordering, switching kinks, and compliance-limited portions that a compact model must actually reproduce. The full cycle ensemble is retained separately and drives both the out-of-sample test (Section IV-C) and the bootstrap (Section III-H).

B. Conduction-Regime Classification

Each branch of the medoid cycle is split at its voltage extremum into pre/post-switching state segments (HRS and LRS sides of SET and RESET). Within sliding windows of each segment, four standard transport linearizations are regressed:

$$\text{ohmic:} \quad \ln I = m \ln V + b, \quad 0.8 \leq m \leq 1.2, \quad (2)$$

$$\text{PF:} \quad \ln(I/V) = s_{\text{PF}} \sqrt{V} + b, \quad s_{\text{PF}} > 0, \quad (3)$$

$$\text{Schottky:} \quad \ln I = s_{\text{S}} \sqrt{V} + b, \quad s_{\text{S}} > 0, \quad (4)$$

$$\text{FN:} \quad \ln(I/V^2) = s_{\text{FN}}/V + b, \quad s_{\text{FN}} < 0. \quad (5)$$

PF and Schottky linearizations are nearly collinear over short windows [13], so a physical admissibility test precedes model selection: the dynamic relative permittivity implied by the fitted $\sqrt{V}$ slope s ,

$$\varepsilon_r^{\text{dyn}} = \frac{q^3}{\pi \varepsilon_0 t_{\text{ox}} (skT)^2} \times \begin{cases} 1, & \text{PF,} \\ 1/4, & \text{Schottky,} \end{cases} \quad (6)$$

must fall in the deliberately broad bracket $[1, 60]$ under the device oxide thickness $t_{\text{ox}} = 7 \text{ nm}$, $T = 300 \text{ K}$. Admitted candidates are compared by the Bayesian information criterion; adjacent same-label windows are merged and refit. Windows labeled ohmic or PF are marked `stanford_valid`, since these are the behaviors the released Stanford current law can emulate; Schottky, FN, and unclassified windows are not. When the cycle ensemble is supplied, every cycle is classified at the state-segment level, yielding a cycle-to-cycle mechanism-stability table.

C. Compact Model and HSPICE Forward Map

The fitted model is the Stanford-PKU RRAM Verilog-A model `rram_v_1_0_0` [4], [5], simulated in HSPICE through generated transient netlists. The terminal current and deterministic gap dynamics are

$$I_{\text{tb}} = I_0 \exp\left(-\frac{g}{g_0}\right) \sinh\left(\frac{V_{\text{tb}}}{V_0}\right), \quad (7)$$

$$\frac{dg}{dt} = -\nu_0 \exp\left(-\frac{qE_a}{kT_{\text{cur}}}\right) \sinh\left(\frac{\gamma a_0}{t_{\text{ox}}} \frac{qV_{\text{tb}}}{kT_{\text{cur}}}\right), \quad (8)$$

with local temperature $T_{\text{cur}} = T_{\text{ini}} + |V_{\text{tb}} I_{\text{tb}} R_{\text{th}}|$ and a gap-dependent field-enhancement factor γ . A series resistor R_s models probe/parasitic resistance. Each objective evaluation runs two transients: a SET sweep initialized in HRS and a RESET sweep initialized in LRS, with polarity-split parameters (V_0 , E_a , F_{min} , ν_0 , g_{max} , g_{ini}) so the model can express measured SET/RESET asymmetry while scale and parasitic parameters remain shared. Measured voltage samples are replayed as a quasi-static piecewise-linear source; the fitted ν_0 and E_a therefore absorb the effective sweep-time scale and should be compared within this protocol rather than read as protocol-independent kinetic constants.

D. Regime-Conditioned Priors

Each of the 21 model parameters receives a prior mean, standard deviation, and hard physical bounds. When a regime map is available, the prior windows are selected from classified physics instead of fixed voltage ranges: the LRS current scale (I_0 , V_0) is regressed as $I = A \sinh(V/V_0)$ over the classified LRS ohmic window, and—when a PF window exists—HRS PF slopes and intercepts map to γ_0 and E_a priors. When no PF window exists for a branch the code does not force a PF fit onto non-PF data; it retains literature-centered defaults ($\gamma_0 = 16.5$, $E_a = 0.6 \text{ eV}$) with inflated uncertainty and records the missing PF prior in a `pf_available` flag. As shown in Section IV-A, this honesty mechanism is essential for the present devices. The series-resistance prior is sized from the measured LRS resistance.

E. Mechanism-Aware Objective

The optimizer minimizes a variance-scaled log-current loss with switching threshold and prior penalties,

$$L(\theta) = L_{\text{SET}} + L_{\text{RESET}} + 0.5L_V + 0.1L_{\text{prior}}, \quad (9)$$

where each branch loss is a weighted mean of squared residuals $r_i = (\log_{10} |I_i^{\text{sim}}| - \log_{10} |I_i^{\text{meas}}|) / \sigma_i^{\log I}$, L_V penalizes the mismatch of switching thresholds located at the extrema of $d \log_{10} |I| / dV$, and the prior term is a quadratic penalty in prior-standardized coordinates. Points covered only by Schottky/FN windows receive weight $w_{\text{nb}} = 0.3$; ohmic/PF and unclassified points receive full weight. The down-weighting limits the ability of structurally unmodeled interface conduction to bias parameters that the Stanford law can represent, without hiding the mismatch.

F. Global Search and Local Refinement

Global search uses Bayesian optimization (expected-improvement-plus, 48 quasi-random seeds + 160 adaptive evaluations, fixed seed) over prior-scaled boxes, with log transforms for scale-like parameters [16], [17]; 16 of the 21 parameters are active by default. The best point is polished by Nelder-Mead in log-parameter coordinates (100 evaluations) with out-of-bounds penalties. A nominal point estimate costs $2(48 + 160 + 100) = 616$ HSPICE branch transients.

G. Operational Identifiability and Generalization

Two complementary, resampling- and sensitivity-based diagnostics are computed at the fitted optimum. *First*, generalization is quantified out of sample: the fitted simulated curve is scored against every measured cycle of the device other than the medoid (Section IV-C); because no non-medoid cycle entered the objective, the resulting RMSE distribution is a held-out test of whether the calibration describes the device rather than the chosen target. *Second*, parameter-level identifiability is read primarily from the cycle bootstrap (Section III-H): a parameter that moves substantially when the measured cycles are resampled is not operationally identifiable, regardless of any local sensitivity. A finite-difference Fisher matrix $F = \tilde{S}^T \tilde{S}$ in log-parameter coordinates, with effective uncertainties $\sigma_j^{\text{eff}} = \sqrt{[F^+]_{jj}}$ from the pseudo-inverse, is retained only as a *local* cross-check [9], [11]. Because the Fisher matrices for this model are extremely ill-conditioned, we use them to characterize the overall sloppiness of the parameterization, not to certify individual coordinates; per-parameter claims are made only where the bootstrap agrees.

H. Cycle-to-Cycle Variability

Cycle-to-cycle uncertainty uses 200 cluster-bootstrap draws over whole measured cycles of the selected device: each draw resamples cycles with replacement, collapses them to one median curve by branch and point index (preserving hysteresis sequence), and refits with a reduced budget (8+24 evaluations) [18]. Percentiles across draws give empirical 95% confidence intervals that reflect cycle-to-cycle variability under the fitting protocol; they make no claim about device-to-device or wafer-level populations.

I. Ablation Design

To isolate the contribution of each ingredient, every condition is also fitted under controlled variants: physics priors with Bayesian optimization (`physics_bo`), uninformative priors over the hard physical bounds (`plain_bo`), prior means without any optimization (`physics_nobo`), and the full regime-conditioned method (`physics_regime_bo`). The ablation reports *unweighted* validation RMSE so that methods using mechanism-aware training weights are not favored by their own loss.

IV. RESULTS

A. Conduction Regimes: Ohmic LRS, Schottky-Limited HRS

Figure 1 shows the classified regime map for the representative S1 device, and Fig. 2 summarizes the cycle-level dominant mechanism for all twelve conditions. The post-SET LRS is ohmic-dominant for every condition, with stability from 66.3% (S2) to 100%. More consequentially, the post-RESET HRS is Schottky-dominant for all conditions in 92.5–100% of cycles. Its representative-cycle Schottky windows imply $\varepsilon_r^{\text{dyn}} = 4.72\text{--}33.96$ under $t_{\text{ox}} = 7\text{ nm}$, inside the deliberately broad admissibility interval.

The pre-SET HRS is Schottky-dominant in ten conditions, with stability from 60.4% (S2) to 100% (S8); S5 and S7 are instead ohmic-dominant at 80% and 75%, respectively. The pre-RESET LRS is the least stable segment and is often unclassified, consistent with a transition between the conductive filament branch and rupture. PF windows occur only in the representative pre-SET maps of S6 and S11, and PF is not the dominant cycle-level mechanism for any state or condition.

This result has two consequences for compact-model extraction. First, the PF-to- (γ_0, E_a) prior pathway fires only on the SET side of S6 and S11; all RESET-side and remaining SET-side activation-energy priors retain literature-centered defaults with inflated uncertainty. Second, a substantial fraction of the HRS sweep is covered only by `stanford_valid=0` windows and enters the loss at weight 0.3. The Stanford current law (7) contains no interface-barrier term, so residual structure in these regions is expected and is quantified below.

B. Point-Estimate Fit Quality

Table II reports unweighted validation RMSE (in decades of log-current) and Durbin–Watson (DW) statistics for all twelve conditions. SET-branch RMSE spans 0.47–0.77 decades (mean 0.530) and RESET-branch RMSE spans 0.65–1.02 decades (mean 0.761). The fitted models capture the hysteresis ordering, both switching transitions, the compliance plateau, and the LRS return branches; the dominant residual is a localized error of up to one decade at the SET transition and along the Schottky-limited HRS approach.

The DW statistics (0.46–1.04, all well below 2) quantify what the overlay plots suggest: residuals are strongly autocorrelated along the sweep. We report this deliberately. Structured residuals indicate systematic compact-model mismatch—consistent with the missing interface-conduction

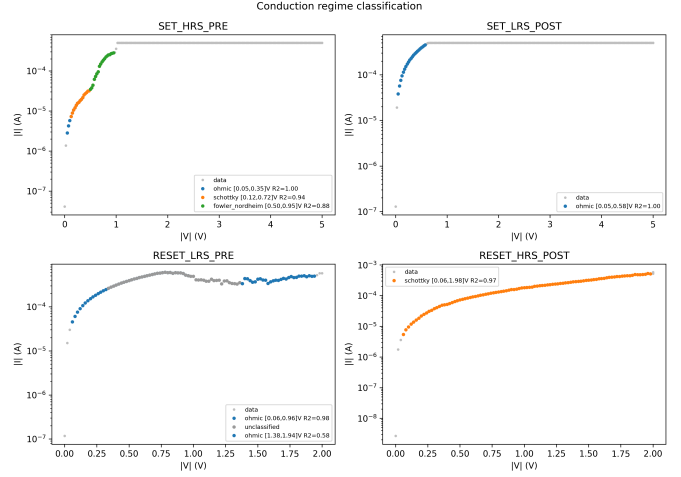


Fig. 1. Conduction-regime classification of the four resistance-state segments of the S1 representative cycle. LRS segments are ohmic ($R^2 \geq 0.98$); HRS segments are dominated by Schottky emission, with a Fowler–Nordheim window near the SET transition. No Poole–Frenkel window is admitted by the $\varepsilon_r^{\text{dyn}}$ test.

TABLE II
VALIDATION FIT QUALITY PER CONDITION: UNWEIGHTED RMSE OF $\log_{10} |I|$ (DECADES) AND DURBIN–WATSON STATISTICS.

Cond.	RMSE _{SET}	RMSE _{RESET}	DW _{SET}	DW _{RESET}
S1	0.500	0.689	0.86	0.85
S2	0.772	1.016	0.46	0.56
S3	0.566	0.659	0.72	0.98
S4	0.514	0.730	0.82	0.90
S5	0.508	0.805	0.85	0.74
S6	0.468	0.650	0.94	0.93
S7	0.477	0.726	1.04	0.91
S8	0.495	1.001	1.03	0.54
S9	0.510	0.700	0.92	1.04
S10	0.508	0.718	1.00	0.98
S11	0.494	0.753	0.92	0.88
S12	0.548	0.681	0.96	1.01
Mean	0.530	0.761	—	—

term identified in Section IV-A—rather than random measurement noise, and averaging them away with looser metrics would overstate model adequacy. S2, with the largest RESET RMSE (1.02 decades) and the lowest SET DW, also has the least stable Schottky assignment among the Schottky-dominant pre-SET conditions (60.4%), supporting this reading.

C. Out-of-Sample Generalization Across Measured Cycles

A point estimate fitted to one medoid cycle is only useful if it also describes the cycles that were *not* used for fitting. We test this directly and without further simulation: the fitted simulated curve is recovered from the archived validation residuals and scored, point by point on the common grid, against every measured cycle of the device. By construction the medoid cycle returns the in-sample RMSE of Table II; the remaining 19–100 cycles per condition are genuinely held out.

Figure 3 reports the resulting per-cycle RMSE distributions with the in-sample value overlaid. The held-out median RMSE, averaged over conditions, is 0.542 decades (SET) and

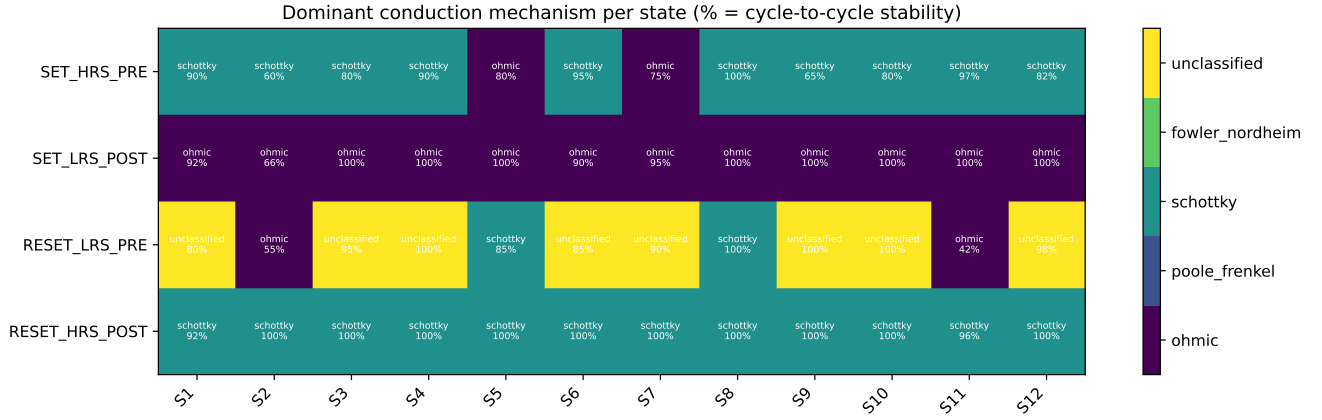


Fig. 2. Dominant conduction mechanism and cycle-to-cycle assignment stability for each resistance-state segment and deposition condition. The post-SET LRS is consistently ohmic, whereas the post-RESET HRS is Schottky-dominant for all twelve conditions.

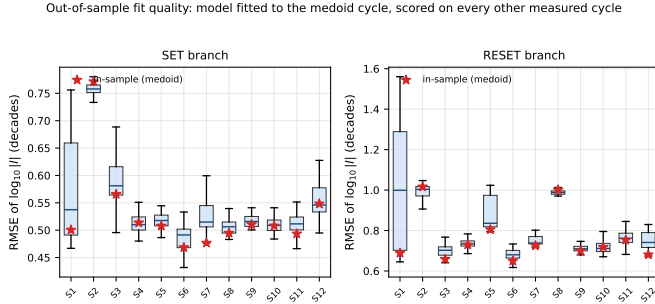


Fig. 3. Out-of-sample fit quality. Box plots: distribution of $\log_{10} |I|$ RMSE when the model fitted to the medoid cycle is scored against every other measured cycle of the device (whiskers exclude outliers). Stars mark the in-sample (medoid) RMSE. Held-out and in-sample errors are statistically indistinguishable, evidencing generalization rather than overfitting.

0.800 decades (RESET), versus in-sample means of 0.530 and 0.761—a generalization gap of only 0.012 and 0.039 decades. For every condition the in-sample marker lies inside the held-out interquartile range, and the held-out 95th percentile exceeds the in-sample value by at most a few hundredths of a decade in the stable conditions. The calibration is therefore a property of the device’s cycle population, not an artifact of medoid selection: the medoid is simply the most central member of a distribution the model reproduces uniformly.

Figure 4 visualizes the test for S1: the fitted curve tracks the measured median and remains inside the cycle-to-cycle envelope along most of the sweep, departing only at the SET transition and the Schottky-limited HRS approach. Consistently, the fitted SET curve falls inside the measured cycle min–max band at 77% of grid points on average across conditions, versus 63% for RESET; the lower RESET coverage and its larger absolute error reflect the *same* structural HRS mismatch identified in Section IV-A, not target overfitting, since both the in-sample and held-out RESET errors are elevated together.

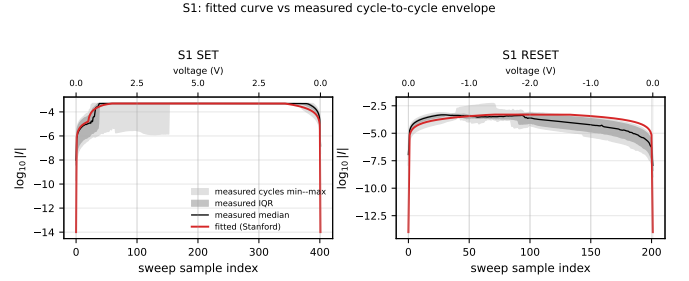


Fig. 4. S1 fitted curve (red) against the measured cycle-to-cycle envelope (min–max and interquartile bands, with the measured median) along the sweep sequence; the secondary axis marks voltage. The fit stays within the measured spread except at the SET transition and the HRS approach.

D. Identifiability: What DC Data Actually Constrain

We read identifiability primarily from the cycle bootstrap. Figure 5 reports the coefficient of variation (CV) of each bootstrap distribution across all twelve conditions; all 2400 refits (200 draws $\times$ 12 conditions) completed and all 192 condition–parameter intervals are finite and nondegenerate. The median CV over all entries is 0.448 (range 0.202–2.698). The branch voltage scales are the most repeatable coordinates (median CV $\approx$ 0.28), consistent with the switching-curve shape directly constraining them. In contrast, series resistance R_s (median CV 1.444) and the SET/RESET velocity prefactors (1.194 and 1.282) are highly unstable, and current scale I_0 and activation energies vary substantially (CV 0.566 and 0.456–0.591).

The Fisher analysis is consistent with—but, on its own, weaker than—this picture, and we present it only as a local cross-check. Across conditions the Fisher condition numbers range from 6.2×10^{18} to 1.6×10^{66} , i.e. far beyond the resolution of double-precision arithmetic; we therefore do *not* certify individual parameters from the pseudo-inverse and read these matrices only as confirmation that the parameterization is extremely sloppy (Fig. 6). Where the local Fisher diagnostic and the bootstrap *agree*—the branch voltage scales and switching-field combinations are comparatively well determined, while I_0 , g_0 , the activation energies, the velocity prefactors, and the

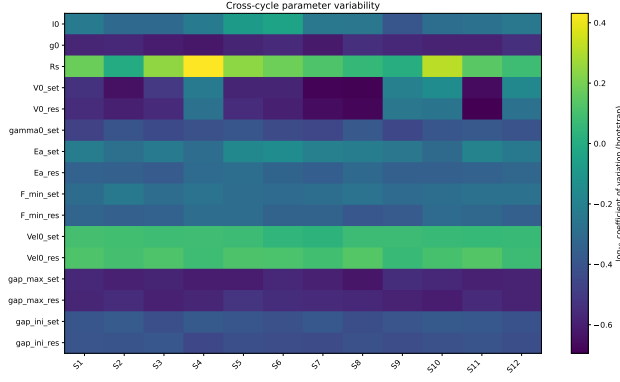


Fig. 5. Cycle-to-cycle parameter variability from 200 cluster-bootstrap refits per condition (color: $\log_{10}$ CV). Branch voltage scales are comparatively repeatable; series resistance and velocity prefactors are the least stable. This bootstrap CV is our primary identifiability evidence.

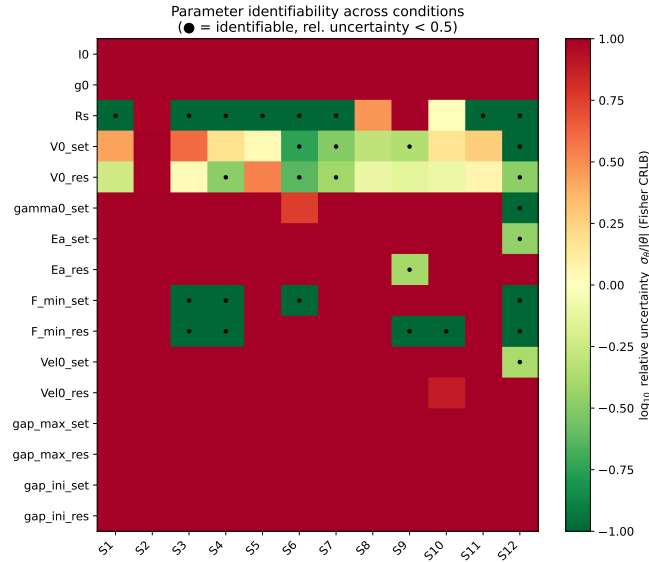


Fig. 6. Local Fisher relative uncertainty $\sigma_{\theta}/|\theta|$ (log color scale) for the 16 active parameters across the twelve conditions, shown as a corroborating cross-check. The extremely large condition numbers (10^{18} – 10^{66}) preclude certifying individual coordinates from this diagnostic alone; conclusions are drawn only where it agrees with Fig. 5.

gap-geometry parameters are not—the conclusion is robust. Where they disagree it is instructive: R_s attains the smallest nominal Fisher relative uncertainty (and is the most recurrent locally identifiable coordinate) yet has the *largest* bootstrap CV, a textbook case of a coordinate that is locally constrained at a single optimum but not repeatable under data resampling. A subset of profile-likelihood scans (S1–S4) independently returns “flat/non-identifiable” for nearly all coordinates, corroborating the bootstrap reading.

The defensible interpretation is that DC butterfly data constrain effective branch scales—voltage curvature and switching-field combinations, and to a lesser and less repeatable extent series resistance—more reliably than the decomposition into I_0 , g_0 , activation energies, velocity prefactors, and gap geometry.

TABLE III
ABLATION OVER ALL 12 CONDITIONS: UNWEIGHTED VALIDATION RMSE (DECADES), MEAN $\pm$ STANDARD DEVIATION ACROSS CONDITIONS.

Variant	RMSE _{SET}	RMSE _{RESET}
Regime-conditioned + weighting	0.530 ± 0.081	0.761 ± 0.123
Physics priors + BO	0.546 ± 0.077	0.756 ± 0.124
Uninformative priors + BO	0.537 ± 0.076	0.755 ± 0.126
Physics priors, no BO	0.571 ± 0.115	1.528 ± 0.434

E. Ablation: What Each Ingredient Buys

Table III compares the extraction variants on unweighted validation RMSE averaged over all twelve conditions. Bayesian optimization is decisive on RESET: physics-informed prior means without optimization yield 1.528 ± 0.434 decades, versus 0.756 ± 0.124 with BO. SET is less sensitive because its LRS and compliance-limited portions are already close to the prior prediction.

The three optimized variants have nearly identical aggregate unweighted error. We do not present regime conditioning or informative priors as an RMSE improvement; their role is to prevent unsupported Schottky/FN windows from steering a bulk-current model and to document how poorly identified coordinates are selected. In a sloppy model, comparable errors from different prior structures are themselves evidence that an overlay is insufficient to establish physical parameter extraction—which is precisely why the generalization and identifiability diagnostics above, not the RMSE, are the deliverable.

F. Deposition-Resolved Extraction

The DOE of Table I is designed to expose parameter trends versus oxygen content and sputter power. Our identifiability results determine *which* parameters such a response surface may legitimately use: only combinations that are both locally constrained and repeatable under the bootstrap. Figure 7 plots the two classes side by side. The branch voltage scales $V_{0,SET}$ and $V_{0,RESET}$ (tight bootstrap CIs, CV ≈ 0.28) vary systematically with deposition—for example $V_{0,RESET}$ tends lower at the 35% oxygen, low-power corners (S3, S4) than at the oxygen-poor corners (S1, S2)—so they can anchor a response surface. Series resistance, by contrast, although the most frequently *locally* identifiable coordinate, carries bootstrap 95% intervals spanning one to two orders of magnitude; its apparent variation across conditions is not separable from cycle-to-cycle noise and must not be read as a deposition trend. This is the practical payoff of separating local identifiability from repeatability: it tells the response-surface user which columns of the extracted parameter table are admissible.

Finally, the regime classifier yields a device-wide, physically interpretable HRS observable. Figure 8 plots the dynamic relative permittivity implied by the post-RESET Schottky window for every condition. All twelve values fall in the physically admissible range 4.7–32, confirming that the dominant HRS conduction is interface-limited Schottky emission across the entire DOE, not a fitting artifact of a single device. The ~ 3 -fold scatter of $\varepsilon_r^{\text{dyn}}$ across the four nominally identical center-point devices (S9–S12) indicates that the interface response

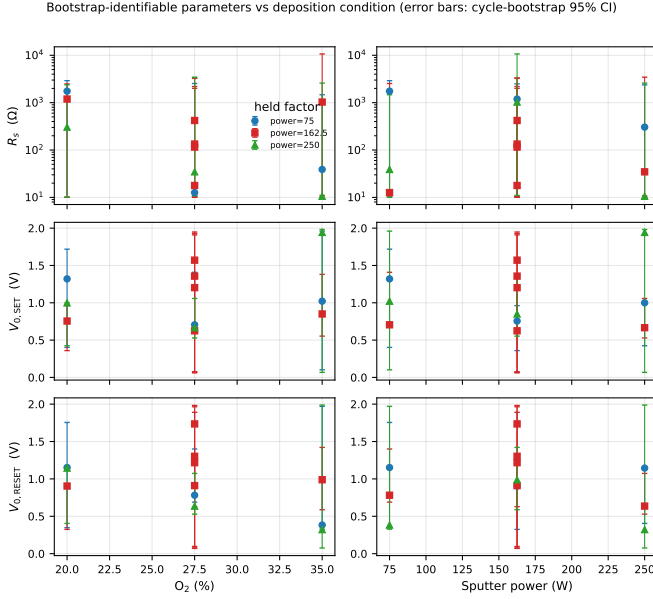


Fig. 7. Bootstrap-identifiable parameters versus deposition condition (error bars: cycle-bootstrap 95% CI). Branch voltage scales (middle, bottom rows) are tightly constrained and carry deposition information; series resistance (top) spans up to two decades and cannot support a response surface despite being locally Fisher-identifiable.

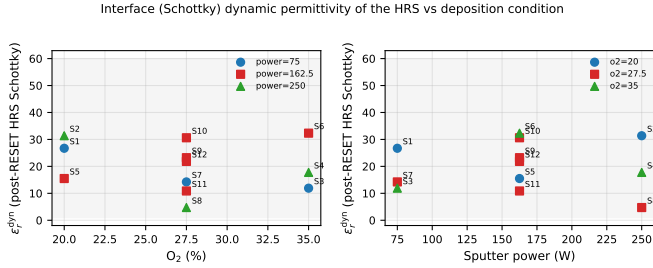


Fig. 8. Dynamic relative permittivity inferred from the post-RESET HRS Schottky window for all twelve conditions, versus oxygen (left) and power (right). Shaded band: admissibility interval [1, 60]. All conditions are physically admissible, corroborating interface-limited HRS conduction device-wide.

itself varies cycle to cycle and device to device, rather than being a tight function of the nominal deposition setpoint—an automated, quantitative version of the mechanism analysis previously performed by hand [19].

V. DISCUSSION

A. Implications for Compact-Model Users

For circuit designers consuming calibrated Stanford parameter sets, the practical messages are direct. (i) The calibration *generalizes*: a parameter set fitted to one representative cycle predicts the other measured cycles of that device with the same error, so per-condition files are usable behavioral models for the fitted protocol. (ii) Only the *identifiable, repeatable* combinations transfer with evidential support; do not ascribe physical meaning to fitted E_a , ν_0 , or gap geometry from DC sweeps alone, and do not trust a coordinate’s small local uncertainty (R_s) without checking its bootstrap repeatability. (iii) When comparing parameters across deposition conditions,

compare the admissible combinations (branch voltage scales, switching thresholds), as in Section IV-F, rather than raw parameter columns that mix data- and prior-driven coordinates.

B. Model Adequacy for TaO_x

The regime analysis localizes the Stanford model’s structural limitation for these devices: HRS conduction is interface-limited (Schottky) in nearly all cycles, while the model’s single sinh current law is bulk-like. This mismatch, not optimizer failure, sets the residual floor of Table II, as confirmed by the near-identical aggregate unweighted RMSE of the three optimized ablation arms *and* by the out-of-sample test, in which the RESET error is elevated identically in and out of sample. A principled extension would add a barrier-limited HRS current branch (or a gap-dependent barrier term) [6], [7], [13]; the regime map produced here provides exactly the windows and slopes such an extension should target. Until then, down-weighting unmodelable windows during fitting prevents interface physics from contaminating bulk parameters while keeping the mismatch visible in unweighted validation.

C. Limitations

The transient replay is quasi-static, so kinetic parameters absorb the effective time scale; constraining kinetics requires pulsed or sweep-rate-resolved data [14]. Identifiability rests on the combined out-of-sample and bootstrap evidence, with Fisher used only as a local cross-check and a partial set of profile-likelihood scans (S1–S4) as corroboration; a future release should complete profile likelihood at one common optimum for all conditions. The bootstrap and out-of-sample analyses are cycle-level for one selected device per condition and therefore quantify cycle-to-cycle, not device-to-device or wafer-level, variability—although the four replicated center-point conditions give a first indication of device-to-device spread. Finally, the reset-side γ_0 is structurally overridden under negative bias in the released Verilog-A, so it is retained for model compatibility but not physically interpreted.

VI. CONCLUSION

We presented an identifiability-aware pipeline that fits the Stanford RRAM compact model to measured TaO_x DC switching data and reports, with every parameter set, the evidence behind it. Applied uniformly to twelve sputter-deposition conditions, the flow achieves mean RMSE of 0.530 (SET) and 0.761 (RESET) decades and establishes findings that overlay plots alone cannot. First, the single-cycle calibration generalizes: scored against all other measured cycles, its held-out median RMSE (0.542/0.800 decades) matches the in-sample value to within 0.04 decades, so the fit reflects the device rather than the chosen target. Second, cycle-bootstrap analysis shows that only the branch voltage scales are repeatable, while series resistance and velocity prefactors are not—and that a coordinate’s small *local* Fisher uncertainty (e.g. R_s) does not imply repeatability. Consequently, only the identifiable, repeatable combinations carry usable deposition information across the DOE. Third, the post-RESET HRS is

Schottky-limited in 92.5–100% of cycles for every condition, with physically admissible permittivities, a mechanism outside the Stanford bulk current law that bounds achievable fidelity. Generalization, identifiability, and mechanism validity should therefore be reported together as the deliverable of RRAM compact-model calibration.

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